

API testing

Start SpringBoot application

POST: <http://localhost:8080/api/rca/search>

Body => raw

```
{
  "question": "Why are payments failing?"
}
```

```
{
  "question": "Why are payments failing?",
  "answer": "Retrieved 2 evidence records from Pinecone",
  "evidence": [
    "ID: trace-TRACE-10046\nScore: 0.1906171292066574\nTrace ID: TRACE-10046\nTransaction ID: TXN-790\nServices: account-service,auth-service,notification-service,payment-service\nError Codes: BALANCE_CHECK_FAILED,DB_CONNECTION_ERROR,DB_POOL_EXHAUSTED\nEnvironment: simulated-prod\n\nEvidence:\nEvidence logs for traceId TRACE-10046 and transactionId TXN-790.\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:17:29Z | service=auth-service | level=INFO | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=105 | message=Token validation successful\n2026-06-22T10:17:30Z | service=account-service | level=ERROR | eventType=APPLICATION_ERROR | errorCode=DB_CONNECTION_ERROR | latencyMs=5100 | message=Account balance lookup failed because database connection timed out\n2026-06-22T10:17:31Z | service=payment-service | level=ERROR | eventType=APPLICATION_ERROR | errorCode=BALANCE_CHECK_FAILED | latencyMs=5200 | message=Payment failed because account balance could not be verified\n2026-06-22T10:17:32Z | service=account-service | level=ERROR | eventType=DATABASE_ERROR | errorCode=DB_POOL_EXHAUSTED | latencyMs=5000 | message=Connection pool exhausted while connecting to account database\n2026-06-22T10:17:33Z | service=notification-service | level=WARN | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=80 | message=Payment failure notification queued for customer\n",
    "ID: trace-TRACE-10045\nScore: 0.13343356549739838\nTrace ID: TRACE-10045\nTransaction ID: TXN-789\nServices: account-service,auth-service,fraud-service,payment-service\nError Codes: FRAUD_TIMEOUT,POD_RESTARTED,SERVICE_UNHEALTHY\nEnvironment: simulated-prod\n\nEvidence:\nEvidence logs for traceId TRACE-10045 and transactionId TXN-789.\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:15:29Z | service=auth-service | level=INFO | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=120 | message=Token validation successful\n2026-06-22T10:15:30Z | service=account-service | level=INFO | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=210 | message=Account balance check successful\n2026-06-22T10:15:31Z | service=payment-service | level=ERROR | eventType=APPLICATION_ERROR | errorCode=FRAUD_TIMEOUT | latencyMs=4300 | message=Payment failed due to fraud-service timeout\n2026-06-22T10:15:33Z | service=fraud-service | level=ERROR | eventType=READINESS_PROBE_FAILED | errorCode=SERVICE_UNHEALTHY | latencyMs=4200 | message=Readiness probe failed: /health returned 503\n2026-06-22T10:15:34Z | service=fraud-service | level=WARN | eventType=POD_RESTARTED | errorCode=POD_RESTARTED | latencyMs=0 | message=fraud-service pod restarted after repeated health check failures\n"
```

```
}  
}
```

POST http://localhost:8080/api/rca/search

Body

```
{  
  "question": "Why are payments failing?"  
}
```

200 OK • 2.61 s • 3.01 KB

JSON

```
{  
  "question": "Why are payments failing?",  
  "answer": "Retrieved 2 evidence records from Pinecone",  
  "evidence": [  
    "ID: trace-TRACE-10046\nScore: 0.1906171292066574\nTrace ID: TRACE-10046\nTransaction ID: TXN-790\nServices: account-service,auth-ser\nnotification-service,payment-service\nError Codes: BALANCE_CHECK_FAILED,DB_CONNECTION_ERROR,DB_POOL_EXHAUSTED\nEnvironment:\nsimulated-prod\nEvidence logs for traceId TRACE-10046 and transactionId TXN-790.\nThe following logs show the request\nflow in timestamp order:\n\n2026-06-22T10:17:29Z | service=auth-service | level=INFO | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=105 | message=Token validation successful\n2026-06-22T10:17:30Z | service=account-service | level=ERROR | eventType=APPLICATION_ERROR | errorCode=DB_CONNECTION_ERROR | latencyMs=5100 | message=Account balance lookup failed because data connection timed out\n2026-06-22T10:17:31Z | service=payment-service | level=ERROR | eventType=APPLICATION_ERROR | errorCode=BALANCE_CHECK_FAILED | latencyMs=5200 | message=Payment failed because account balance could not be verified\n2026-06-22T10:17:32Z | service=account-service | level=ERROR | eventType=DATABASE_ERROR | errorCode=DB_POOL_EXHAUSTED | latencyMs=5000 | message=Connection pool exhausted while connecting to account database\n2026-06-22T10:17:33Z | service=notification-service | level=WARN | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=80 | message=Payment failure"
```

POST http://localhost:8080/api/rca/analyze

Body => raw

```
{  
  "question": "Why are payments failing?"  
}
```

```
{  
  "question": "Why are payments failing?",  
  "answer": "1. **Root Cause** \n  The primary cause of the payment failures is a combination of  
database connection issues in the account-service and timeout errors with the fraud-service.  
Specifically, the account-service encountered a database connection timeout and an exhausted  
connection pool, while the fraud-service was unhealthy and timed out during payment processing.\n\n2.  
**Symptom** \n  Payments are failing for users, resulting in error messages indicating that account  
balance verification could not be completed and that there were issues with fraud detection.\n\n3.  
**Impacted Service** \n  The impacted service is the payment-service, which relies on both the  
account-service for balance verification and the fraud-service for fraud checks.\n\n4. **Evidence** \n
```

- For Transaction ID TXN-790 (TRACE-10046): \n - 2026-06-22T10:17:30Z | account-service | ERROR | DB_CONNECTION_ERROR | Account balance lookup failed because database connection timed out. \n - 2026-06-22T10:17:31Z | payment-service | ERROR | BALANCE_CHECK_FAILED | Payment failed because account balance could not be verified. \n - 2026-06-22T10:17:32Z | account-service | ERROR | DB_POOL_EXHAUSTED | Connection pool exhausted while connecting to account database. \n - For Transaction ID TXN-789 (TRACE-10045): \n - 2026-06-22T10:15:31Z | payment-service | ERROR | FRAUD_TIMEOUT | Payment failed due to fraud-service timeout. \n - 2026-06-22T10:15:34Z | fraud-service | ERROR | SERVICE_UNHEALTHY | Readiness probe failed: /health returned 503. \n\n5. ****Suggested Fix**** \n - Investigate and resolve the database connection issues in the account-service, including increasing the connection pool size to prevent exhaustion and ensuring database availability.\n - Check the health and performance of the fraud-service to ensure it can respond in a timely manner and avoid timeouts during payment processing. Consider scaling the service if it's under heavy load or has insufficient resources.\n - Monitor the services closely after fixes are applied to ensure payment processing stabilizes.",

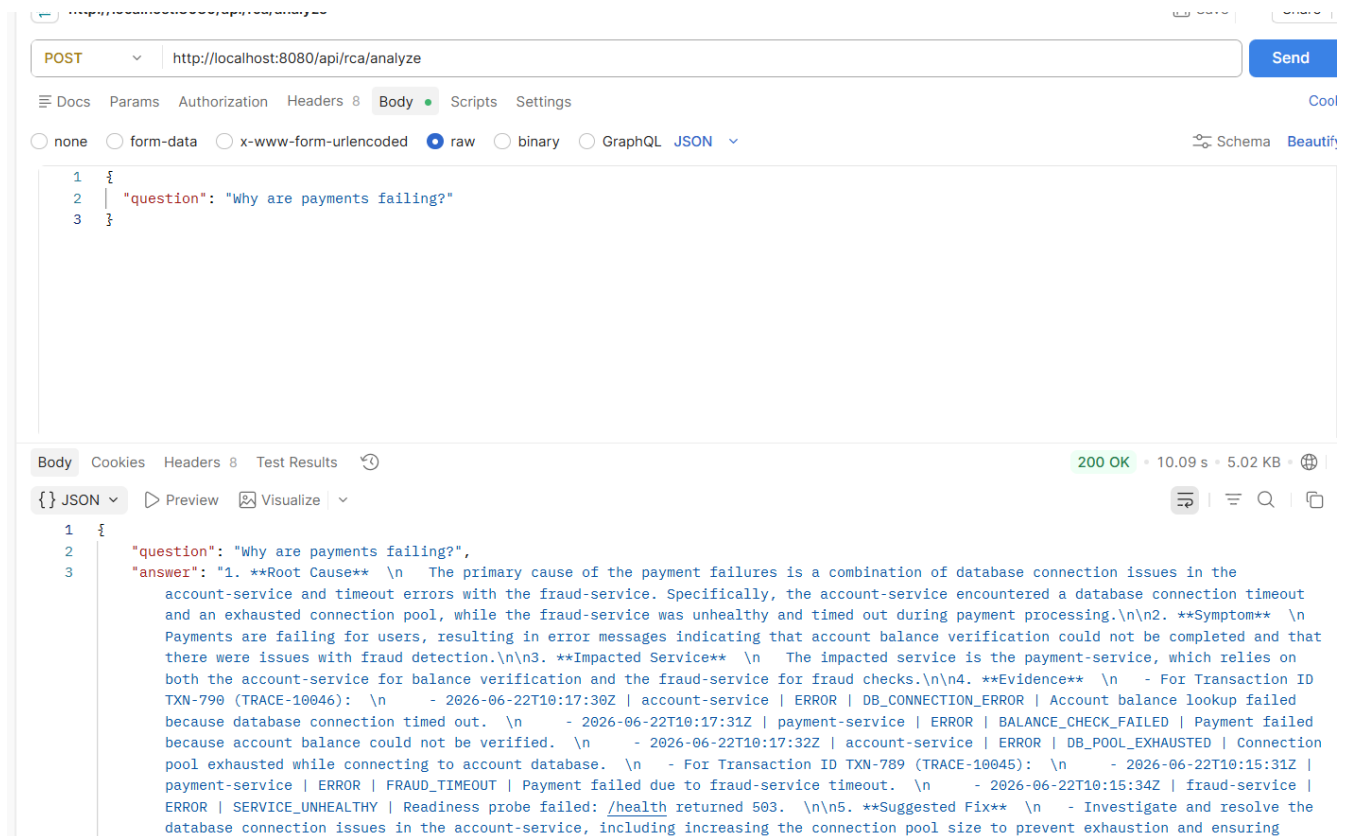
"evidence": [

"ID: trace-TRACE-10046\nScore: 0.1906171292066574\nTrace ID: TRACE-10046\nTransaction ID: TXN-790\nServices: account-service,auth-service,notification-service,payment-service\nError Codes: BALANCE_CHECK_FAILED,DB_CONNECTION_ERROR,DB_POOL_EXHAUSTED\nEnvironment: simulated-prod\n\nEvidence:\nEvidence logs for traceId TRACE-10046 and transactionId TXN-790.\n\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:17:29Z | service=auth-service | level=INFO | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=105 | message=Token validation successful\n2026-06-22T10:17:30Z | service=account-service | level=ERROR | eventType=APPLICATION_ERROR | errorCode=DB_CONNECTION_ERROR | latencyMs=5100 | message=Account balance lookup failed because database connection timed out\n2026-06-22T10:17:31Z | service=payment-service | level=ERROR | eventType=APPLICATION_ERROR | errorCode=BALANCE_CHECK_FAILED | latencyMs=5200 | message=Payment failed because account balance could not be verified\n2026-06-22T10:17:32Z | service=account-service | level=ERROR | eventType=DATABASE_ERROR | errorCode=DB_POOL_EXHAUSTED | latencyMs=5000 | message=Connection pool exhausted while connecting to account database\n2026-06-22T10:17:33Z | service=notification-service | level=WARN | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=80 | message=Payment failure notification queued for customer\n",

"ID: trace-TRACE-10045\nScore: 0.13343356549739838\nTrace ID: TRACE-10045\nTransaction ID: TXN-789\nServices: account-service,auth-service,fraud-service,payment-service\nError Codes: FRAUD_TIMEOUT,POD_RESTARTED,SERVICE_UNHEALTHY\nEnvironment: simulated-prod\n\nEvidence:\nEvidence logs for traceId TRACE-10045 and transactionId TXN-789.\n\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:15:29Z | service=auth-service | level=INFO | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=120 | message=Token validation successful\n2026-06-22T10:15:30Z | service=account-service | level=INFO | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=210 | message=Account balance check successful\n2026-06-22T10:15:31Z | service=payment-service | level=ERROR | eventType=APPLICATION_ERROR | errorCode=FRAUD_TIMEOUT | latencyMs=4300 | message=Payment failed due to fraud-service timeout\n2026-06-22T10:15:33Z | service=fraud-service | level=ERROR | eventType=READINESS_PROBE_FAILED | errorCode=SERVICE_UNHEALTHY | latencyMs=4200 | message=Readiness probe failed: /health returned 503\n2026-06-22T10:15:34Z | service=fraud-service | level=WARN | eventType=POD_RESTARTED | errorCode=POD_RESTARTED | latencyMs=0 | message=fraud-service pod restarted after repeated

health check failures\n"

```
]
}
```



POST http://localhost:8080/api/rca/analyze

Body => JSON

```
{
  "question": "Why did TRACE-10045 fail?"
}
```

```
{
  "question": "Why did TRACE-10045 fail?"
}
```

```
{
  "question": "Why did TRACE-10045 fail?",
  "rootCause": "The fraud-service timed out due to being unhealthy and restarted.",
  "symptom": "Payment failed due to fraud-service timeout.",
  "impactedService": "payment-service",
  "evidence": [
    "2026-06-22T10:15:31Z | service=payment-service | level=ERROR |
```

```
eventType=APPLICATION_ERROR | errorCode=FRAUD_TIMEOUT | latencyMs=4300 |
message=Payment failed due to fraud-service timeout",
    "2026-06-22T10:15:33Z | service=fraud-service | level=ERROR |
eventType=READINESS_PROBE_FAILED | errorCode=SERVICE_UNHEALTHY | latencyMs=4200
| message=Readiness probe failed: /health returned 503",
    "2026-06-22T10:15:34Z | service=fraud-service | level=WARN | eventType=POD_RESTARTED |
errorCode=POD_RESTARTED | latencyMs=0 | message=fraud-service pod restarted after repeated
health check failures"
  ],
  "suggestedFix": [
    "Investigate the health of the fraud-service to prevent timeouts.",
    "Monitor and optimize the readiness probe settings for the fraud-service."
  ],
  "retrievedEvidence": [
    "ID: trace-TRACE-10046\nScore: 0.20818598568439484\nTrace ID:
TRACE-10046\nTransaction ID: TXN-790\nServices: account-service,auth-service,notification-
service,payment-service\nError Codes:
BALANCE_CHECK_FAILED,DB_CONNECTION_ERROR,DB_POOL_EXHAUSTED\nEnvironme
nt: simulated-prod\n\nEvidence:\nEvidence logs for traceId TRACE-10046 and transactionId
TXN-790.\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:17:29Z |
service=auth-service | level=INFO | eventType=APPLICATION_LOG | errorCode=NONE |
latencyMs=105 | message=Token validation successful\n2026-06-22T10:17:30Z | service=account-
service | level=ERROR | eventType=APPLICATION_ERROR |
errorCode=DB_CONNECTION_ERROR | latencyMs=5100 | message=Account balance lookup failed
because database connection timed out\n2026-06-22T10:17:31Z | service=payment-service |
level=ERROR | eventType=APPLICATION_ERROR | errorCode=BALANCE_CHECK_FAILED |
latencyMs=5200 | message=Payment failed because account balance could not be verified\n2026-06-
22T10:17:32Z | service=account-service | level=ERROR | eventType=DATABASE_ERROR |
errorCode=DB_POOL_EXHAUSTED | latencyMs=5000 | message=Connection pool exhausted while
connecting to account database\n2026-06-22T10:17:33Z | service=notification-service | level=WARN |
eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=80 | message=Payment failure
notification queued for customer\n",
    "ID: trace-TRACE-10045\nScore: 0.19158847630023956\nTrace ID:
TRACE-10045\nTransaction ID: TXN-789\nServices: account-service,auth-service,fraud-
service,payment-service\nError Codes:
FRAUD_TIMEOUT,POD_RESTARTED,SERVICE_UNHEALTHY\nEnvironment: simulated-
prod\n\nEvidence:\nEvidence logs for traceId TRACE-10045 and transactionId TXN-789.\nThe
following logs show the request flow in timestamp order:\n\n2026-06-22T10:15:29Z | service=auth-
service | level=INFO | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=120 |
message=Token validation successful\n2026-06-22T10:15:30Z | service=account-service | level=INFO
| eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=210 | message=Account balance
check successful\n2026-06-22T10:15:31Z | service=payment-service | level=ERROR |
eventType=APPLICATION_ERROR | errorCode=FRAUD_TIMEOUT | latencyMs=4300 |
message=Payment failed due to fraud-service timeout\n2026-06-22T10:15:33Z | service=fraud-service
| level=ERROR | eventType=READINESS_PROBE_FAILED | errorCode=SERVICE_UNHEALTHY |
latencyMs=4200 | message=Readiness probe failed: /health returned 503\n2026-06-22T10:15:34Z |
service=fraud-service | level=WARN | eventType=POD_RESTARTED |
errorCode=POD_RESTARTED | latencyMs=0 | message=fraud-service pod restarted after repeated
health check failures\n"
```

```
}
]
```

REST client interface showing a POST request to `http://localhost:8080/api/rca/analyze`. The request body is a JSON object:

```
{
  "question": "Why did TRACE-10045 fail?"
}
```

The response is a 200 OK status with a detailed JSON body:

```
{
  "question": "Why did TRACE-10045 fail?",
  "rootCause": "The fraud-service timed out due to being unhealthy and restarted.",
  "symptom": "Payment failed due to fraud-service timeout.",
  "impactedService": "payment-service",
  "evidence": [
    "2026-06-22T10:15:31Z | service=payment-service | level=ERROR | eventType=APPLICATION_ERROR | errorCode=FRAUD_TIMEOUT | latencyMs=4300 | message=Payment failed due to fraud-service timeout",
    "2026-06-22T10:15:33Z | service=fraud-service | level=ERROR | eventType=READINESS_PROBE_FAILED | errorCode=SERVICE_UNHEALTHY | latencyMs=4200 | message=Readiness probe failed: /health returned 503",
    "2026-06-22T10:15:34Z | service=fraud-service | level=WARN | eventType=POD_RESTARTED | errorCode=POD_RESTARTED | latencyMs=0 | message=fraud-service pod restarted after repeated health check failures"
  ],
  "suggestedFix": [
    "Investigate the health of the fraud-service to prevent timeouts.",
    "Monitor and optimize the readiness probe settings for the fraud-service."
  ],
  "retrievedEvidence": [
    "ID: trace-TRACE-10046\nScore: 0.20818598568439484\nTrace ID: TRACE-10046\nTransaction ID: TXN-790\nServices: account-service, auth-service, notification-service, payment-service\nError Codes: BALANCE_CHECK_FAILED, DB_CONNECTION_ERROR, DB_POOL_EXHAUSTED\nEnvironment: simulated-prod\nEvidence: \nEvidence logs for traceId TRACE-10046 and transactionId TXN-790.\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:17:29Z | service=auth-service | level=INFO |"
  ]
}
```

GET `http://localhost:8080/api/dashboard/overview`

```
{
  "errorsByService": [
    {
      "service": "payment-service",
      "errors": 2
    },
    {
      "service": "fraud-service",
      "errors": 2
    },
    {
      "service": "account-service",
      "errors": 2
    }
  ],
  "errorTrend": [
    {
```

```
    "time": "10:15:31",
    "errors": 1
  },
  {
    "time": "10:15:33",
    "errors": 1
  },
  {
    "time": "10:15:34",
    "errors": 1
  },
  {
    "time": "10:17:30",
    "errors": 1
  },
  {
    "time": "10:17:31",
    "errors": 1
  },
  {
    "time": "10:17:32",
    "errors": 1
  }
],
"errorCodes": [
  {
    "errorCode": "FRAUD_TIMEOUT",
    "count": 1
  },
  {
    "errorCode": "SERVICE_UNHEALTHY",
    "count": 1
  },
  {
    "errorCode": "POD_RESTARTED",
    "count": 1
  },
  {
    "errorCode": "DB_CONNECTION_ERROR",
    "count": 1
  },
  {
    "errorCode": "BALANCE_CHECK_FAILED",
    "count": 1
  },
  {
    "errorCode": "DB_POOL_EXHAUSTED",
    "count": 1
  }
]
```

```

],
"traces": [
  {
    "traceId": "TRACE-10045",
    "transactionId": "TXN-789",
    "services": "payment-service, fraud-service",
    "errorCodes": "FRAUD_TIMEOUT, SERVICE_UNHEALTHY, POD_RESTARTED",
    "severity": "High"
  },
  {
    "traceId": "TRACE-10046",
    "transactionId": "TXN-790",
    "services": "account-service, payment-service",
    "errorCodes": "DB_CONNECTION_ERROR, BALANCE_CHECK_FAILED,
DB_POOL_EXHAUSTED",
    "severity": "High"
  }
]
}

```

GET

▼

http://localhost:8080/api/dashboard/overview

☰ Docs

Params

Authorization

Headers 6

Body

Scripts

Settings

Query Params

Key	Value	Description
Key	Value	Description

BodyCookiesHeaders 8Test Results200 OK • 422 ms • 1.2.1

{ } JSON ▾

PreviewVisualize ▾

1 {

2 "errorsByService": [

3 {

4 "service": "payment-service",

5 "errors": 2

6 },

7 {

8 "service": "fraud-service",

9 "errors": 2

10 },

11 {

12 "service": "account-service",

13 "errors": 2

14 }

15],

16 "errorTrend": [

17 {

18 "time": "10:15:31",

19 "errors": 1

20 }

